AI-based drop-out prediction and counseling system

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Abstract- The AI-powered Drop-out Prediction and Counseling System is created so that it can assist an educational establishment to notice those students that will soon drop their studies. In most colleges, most important academic records like attendance records, internal marks, subject attempts, and fee status are saved in different files or spreadsheet systems. The resulting fragmented data management also makes it challenging to identify signs of early warning about poor attendance or poor performance academically, among faculty. This has led to intervention taking place late when there are poor results in semesters and hence less effective intervention.

The suggested system will bring together the disperse academic records into a centralized SQLite database and operate them with Python and Pandas to analyze them structurally. An evaluation of various performances indicators is conducted by applying a rule-based and a threshold-based lightweight machine learning approach. The system calculates a risk level based on a set of pre-established academic levels, and the students are grouped into the low, medium, and high-risk groups. This will give it clarity, simplicity and easy deployment without complex predictive models. Streamlit is used to create the application, which offers an interactive and user-friendly web interface, with FastAPI and Uvicorn being the basis of the backend operations and risk assessment logic. The system produces colour-coded alerts and dashboard visualisations to enable mentors and counsellors to quickly spot at risk students and promptly either academic or counselling intervention.

The platform allows uploading of data in CSV and spreadsheets, which makes it compatible with the current institutional records. The unified data surveillance coupled with easy AI-based risk classification will make the system a low cost, easily scalable, and convenient solution to proactive student support. It provides teachers with clear insight that is data-driven and at the same time is easy to implement which eventually leads to higher student retention rates and decreased rate of dropouts.

Keywords: Artificial Intelligence (AI), Drop-out Prediction, Student Retention, Counseling Support System, Rule-Based Machine Learning, Threshold-Based Classification, Risk Assessment, Academic Performance Monitoring, Attendance Analysis, Unified Student Dashboard, Streamlit, FastAPI, SQLite, Pandas, Educational Data Integration, Early Warning System.

I. INTRODUCTION

Student dropout is one of the key issues in the higher education institutions and in the public colleges where academic monitoring is mostly done manually. Such early warning signs as decreased attendance rates, low internal assessment scores, repetitive attempts on subjects, and inconsistent fee status are usually saved in different files or spreadsheet systems. Since this data is not centrally integrated, the faculty members cannot trace the trends of academic struggle in time. This leads to intervention being reactive, instead of proactive and most of the students lose focus until it is too late and they are not assisted by getting guidance or counselling.

The AI-powered Drop-out Prediction and Counseling System can resolve this problem with the help of a single and systematic method of academic tracking. The system brings together the attendance records, internal marks, subject attempts and other associated academic information under a centralized SQLite database. The platform is able to provide consistency and accuracy in the analysis of student records by preprocessing and handling the data using Python and Pandas.

The system uses rule-based and threshold-based lightweight machine learning logic to compare various academic indicators. They do not use complex predictive models to classify students as low-risk, medium-risk, and high-risk by using predefined academic thresholds. The simplicity of this AI-based solution is that the system is easy to introduce, comprehend, and manage and at the same time deliver valuable early- warning information.

Streamlit is used as the frontend interface to develop the app, which provides an interactive and user-friendly dashboard that allows the mentors and administrators to interact. FastAPI, which is supported by Uvicorn, controls the logic of processing backends and risk assessment. The system offers risk-levels in colour-codes and organised dashboard outputs that enable the members of the faculty to swiftly single out vulnerable students and start the process of counselling or academic assistance in a timely manner.

The system encourages the use of data to make decisions by combining the existing institutional data into one monitoring system and implementing a structured process of risk classification without having to install costly analytics software. The proposed solution will focus on facilitating the early intervention, enhance student engagement, and decrease dropout rates with the help of proactive academic counselling and constant monitoring.

II. METHODOLOGY

The smart drop- out prediction and counseling system is AI-based and has a step-by step systematic approach to the development of the complete system, proper identification of risks, and application to the institutions. The methodology

is separated into four big steps namely: Data Integration, Risk Evaluation, System Implementation, and Monitoring and Intervention. The stages are critical in the development of an effective and light based academic monitoring system.

Phase 1: Data Collection and Integration: The initial phase is aimed at the gathering and sorting of student academic records into a unified system. The attendance information, internal assessment marks, subject attempts, and fee status are stored separately in Excel spreadsheets or files in most institutions. In this dismemberment, it becomes hard to see coherent academic patterns.
In order to overcome this fragmentation:

- Student record is loaded into the system in CSV or excel files.
- Streamlit frontend is an interface to allow data upload and preview.
- Pandas is applied to clean, format, validate and preprocessor data.
- Missing values, irregular entries and formatting error are processed in the preprocessing stage.
- The analysis information is held in centralized SQLite database.

The database schema is designed such that it will have a well-organized student information and academic indicators. This centralized storage is used to make all the relevant performance parameters available at one platform, as opposed to the scattered records management.

Phase 2: Risk Analysis and Classification: At the second phase, the student performance is evaluated with the rule-based and threshold-based lightweight machine learning logic. In its place of complicated predictive modelling, the system uses preset academic threshold levels to identify the degree of risk.

The criteria of evaluation are:

- The attendance rate low below a set limit.
- Low marks on the internal academic standards.
- Several attempts of the subject that point to recurrent scholastic difficulty.
- Fee anomalies (when incorporated in institutional monitoring).

The indicators are cumulative risk evaluations. The system of classification is based on organized conditional rules. For example:

- Below threshold in terms of attendance and internal marks the risk is Medium.
- Below acceptable limits of attendance, marks and repeated attempts, etc: High Risk.
- When the performance indicators are in acceptable levels, then there is Low Risk.

On these grounds, students are divided into:

- Low Risk
- Medium Risk
- High Risk

This will make faculty members find it easy to understand, interpret their results and have transparency. The logic is rule based and is thus, simple, lightweight and computationally efficient.

Phase 3: System Architecture and Implementation The third stage is on application development and system integration. The building comprises the following elements:

Frontend Layer:

- Developed using Streamlit.
- Offers administration/mentor interactive dashboard.
- Shows student data, category of risks and visual summaries.
- Supports CSV/Excel file import and real time results visualization.

Backend Layer:

- Implemented using FastAPI.
- Uvicorn is the ASGI server to receive API requests.
- Implements risk classifying logic.
- Deals with frontend to database communication.

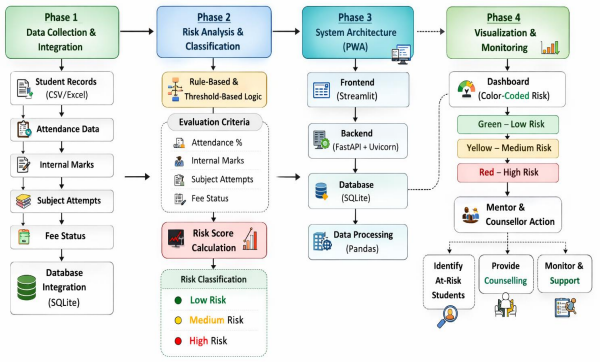
Database Layer:

- SQLite is employed in the central storage of data.
- Keeps student records, returned results and risk.
- Assures portability and low cost of deployment.

Data Processing Layer:

- Pandas is used in structured computation and transformation of data.
- Carries out comparisons against threshold and risk classification.
- Assures of effective management of structured data.

Such a layered architecture equates to modularity, maintainability and future scalability.



Methodology Phases – AI-Based Drop-out Prediction & Counseling System

III. IMPLEMENTATION

The Drop-out Prediction and Counseling System is an AI-based software prototype that was created with the aim of combining academic history of students, conducting systematic risk classification, and offering monitoring capabilities to mentors and administration in a dashboard format. The implementation is focused on the simplicity, modularity, and low-computational needs. The system is built strictly based on Streamlit as the front-end, FastAPI and Uvicorn as the processor of the back-end, SQLite to work with the database, and Pandas to process the structured data. The system development environment and configuration are normal.

A. System Development Environment and Configuration:

Visual Studio Code was used as the main Integrated Development Environment (IDE) in the creation of the system. Python was the main programming language, and Python Virtual Environment (venv) was set up to guarantee the controlled dependency management and package isolation.

The environment was designed in a way that frontend, backend, and database were separated into components that are easier to maintain and have a modular architecture. Components were tested with real-time interaction locally on the system to develop the system.

- **Frontend Development Environment:** Streamlit framework was employed to make an interactive web-based environment. The frontend consists of the modules of uploading CSV / Excel files, visualizing the processed student records, and presenting the results of the risk classification process. The interface was created to be user-friendly and not technical to non-technical users including mentors and faculty members.
- **Backend Development Environment:** Backend routing and risk assessment logic were implemented in FastAPI. Uvicorn is the ASGI server that is used to run FastAPI and process requests effectively. The test of API endpoints was performed in order to make sure that the frontend and backend modules communicate properly.
- **Testing Configuration:** The program was tested on a local development environment. Tests in functional area were to check file uploads, database insertion, logic in rule-based classification, and accuracy of dashboard output. The performance tests made sure that the risk assessment and dashboard representation were responsive despite the presence of numerous student records.

B. Database set up and assimilation

SQLite is the centralized database in which the system stores academic records of the students and computed risk classification. SQLite was used because of its lightweight design, portability, and the little configuration.

- **Central Database (SQLite):** The schema of the database comprised structured variables like percent attendance, inner marks, subject attempts, fee status, and risk category calculated.
- **Storage and Retrieval of Data:** The student records uploaded on the frontend are processed to get a Pandas object which is later loaded into the SQLite tables. There was testing of CRUD operations (Create, Read, Update, Delete) so that there was a reliable interaction with the database.
- **Data Consistency Testing:** esting was done on the recurring upload and update operations to ensure data integrity. The system makes sure that the updated student information is reflected in the dashboard without duplication and inconsistency.

C. Risk Classification and Processing Logic

The fundamental strength of the system is lightweight machine learning logic, which is rule based and threshold based. Rather than complicated predictive modeling, pre-established academic thresholds measure the risk of dropout.

Data Processing with Pandas: Pandas is an analytical tool that helps to purify, format, and validate uploaded data. The missing values are addressed, the numeric formats are checked, and the corresponding academic indicators are obtained to be evaluated.

Rule-Based Risk Evaluation: The system uses academic thresholds that have been predetermine:

- Missing of work more than a certain percentage.
- Lower academic ratings than the minimum requirements.
- Repeated difficulty suggested by several attempts of subject.
- Fee irregularity as another parameter of monitoring.

All the conditions play a role in the general risk assessment. Students are classified into: based on cumulative evaluation:

- Low Risk
- Medium Risk
- High Risk

Openness of Classification: The rule-based framework makes the decisions on risks transparent and understandable by the faculty members without the complexity of black-box models.

D. Analysis and Management Reporting.

Streamlit dashboard is the key monitoring tool of administrators and mentors. The interface displays processed results in a systematic and visually best understandable format.Risk indicators are colour-coded to facilitate easy identification of the risk mitigation process.

- **Colour-Coded Risk Indicators:**
  - Green – Low Risk
  - Yellow – Medium Risk
  - Red – High Risk

This visualisation is based on colour and can be identified quickly. of pupils who demand attention at this time.

- **Interactive Monitoring:** The dashboard will allow the faculty to:
  - View fully integrated student records
  - Keep a track of characterized risk levels
  - Determine academic trends with a number of indicators
- **Usability Focus:** The interface design will be designed with focus on clarity, simplicity and simplicity. The system does not demand technical skills and thus mentors are able to concentrate on counselling as opposed to complexity of the system.

E. System Deployment and Scalability Requirements

The system is already implemented in the local environment to demonstrate and evaluate it. The modular architecture can be extended to cloud platforms like AWS, Azure or GCP without any significant structural changes in the future.

Streamlit, FastAPI, SQLite, and Pandas are lightweight, which means that the solution can be scaled to be hardware-dependent and thus can be used in the institutions with low technical infrastructure..

Implementation Summary:

The introduction of Drop-out Prediction and Counseling System based on AI shows that Streamlit, FastAPI, Uvicorn, SQLite, and Pandas could be successfully combined into a single and functional prototype. The system is efficient in consolidating the scattered academic records into a central database and implements a structured use of rules based classification to overcome the identification of the at-risk students. A mentor and an administrator can easily see the risk levels of students in a visualized form, through the Streamlit dashboard, using colours so that it can be easily understood and interpreted.

The fastapi based modular backend architecture ensures communication between data processing and presentation layers is smooth and the lightweight SQLite offers suitable and dependable data storage. By using Pandas, it is possible to process academic indicators and evaluate them effectively and efficiently without a complicated machine learning model. In general, the developed prototype fulfils its goal, which is to offer a practical, inexpensive, and scalable response to the issue of early dropout identification and counselling service in institutions.

IV. RESULTS AND EVALUATION

The Drop-out Prediction and Counseling System created through AI was a working prototype implemented and tested in one local setting. The assessment is done with regard to confirming the functional accuracy, risk classification performance, system usability and overall reliability. Seeing how the system is developed as a lightweight method of institutional monitoring, the evaluation approach is focused on the proper data integration, the classification logic that relies on rules instead of sophisticated predictive benchmarking.

A. Functional Validation

The system was experimented to make sure that all the core modules are functioning as per the objectives set.

- **Data Upload and Integration:** The spreadsheet-based student records uploaded in the form of CSV/Excel files were manipulated successfully with the help of Pandas and saved in the SQLite database. Structured data like attendance, internal marks, subject attempts and fee status were properly processed by the system.
- **BackendProcessing:** The endpoints of Fast API were checked to validate the proper implementation of rule logic of classification. The backend was able to calculate the levels of risk and deliver results to the Streamlit frontend without rejected communications.
- **The accuracy of the risk classification:** The threshold-based logic was able to classify students into Low, Medium, and High-risk levels correctly in accordance with preset academic conditions. Testing was repeated on different datasets and showed a consistent output in classification.
- **Dashboard Visualization:** Integrated records and colour-coded alerts were correctly presented in the Streamlit dashboard. It was easy to interpret the risk levels as they were depicted visually by Green, Yellow, and Red indicators.

Functional validation is used to verify the reliability of the system in the given scope of the project to carry out its intended functions.

B. Responsiveness and Performance of the System

The system was tested in terms of performance in the local deployment conditions.

- **Processing Efficiency:** As the system is based on rule-based assessment rather than on complicated predictive modeling, the risk is calculated quickly and with low computing energy.
- **Database Performance:** SQLite was an effective data storage and data retrieval system. CRUD was put to test in order to verify that data is updated and read consistently without being duplicated or delayed.
- **Frontend Responsiveness:** Processing results were presented on the Streamlit interface immediately after uploading the files. The dashboard was responsive when it was subjected to repetitive classification processes.

The analysis shows that the system has steady performance with relatively high institutional data loads.

C. User-Friendly System Design

- Streamlit interface has a well-organized workflow structure that enables the mentors and administrations to post student records and access risk classifications with ease without technical challenges.
- Academic attention is provided to students in need with the help of colour-coded risk indicators (Green, Yellow, Red) that allow identifying such students almost instantly, and monitoring them is more efficient and practical.
- The consolidated dashboard incorporates the attendance, marks, attempts of subjects, and fee status into a single platform, decreasing the need to track them manually and taking a closer role in timely intervention counselling.
- The classification method on the basis of rules provides the transparency of the decision-making process, where members of the faculty have a clear vision of the contribution of academic indicators to the risk level of each of the students.

D Evaluation Criteria.

The prototype was tested against the following criteria:

1.Functionality Check: Checking that files uploaded correctly, that they are stored in the database, that there is communication with the API and that the risks are classified
2.Usability Testing: The test of the dashboard intelligibility and monitoring simplicity by non-technical users.
3.Scalability Assessment: To verify consistency of behaviour of the system, repeated data uploads and classification cycles are tested.
Its findings suggest that the prototype introduced is fulfilling the specified goals of student records integration, the use of simple AI logic, and the ability to do structured monitoring.

Summary:
The system created proves to be an effective application of a unified student monitoring platform with the help of Streamlit, FastAPI, SQLite, and Pandas. The assessment is valid when it comes to proper rule-based risk classification, valid database integration, and successful dashboard visualization. Although large-scale implementation and real-time testing on institutional level are only going to be steps into the future, the present prototype has laid out a good groundwork of initial dropout detection and counselling assistance using a feasible and scalable method:

V. CONCLUSION AND FUTURE

A. Conclusion

The Drop-out Prediction and Counseling System is an AI-implemented system that suggests a more direct way of resolving the encountered problem of student dropout by identifying and tracking it down during early stages of their education. The system effectively integrates disconnected academic information, such as attendance records, internal assessment marks, attempts in subjects and payment status into a centralized SQLite database. The system achieves this through a combination of these parameters into one platform, thus removing manual cross-verification and reliance on spreadsheets that are scattered across.

Its application is purely based on a lightweight architecture based on Streamlit to view the interactive dashboard, FastAPI and Uvicorn to process requests, SQLite to store data, and Pandas to work with structured data and calculate risks based on rules. The threshold-based classification mechanism is transparent, interpretable and simplified in determining the Low, Medium and High-risk students. The visualization with colour-coding enables clarity, which will enable the mentor and the administrator to identify vulnerable students in time and provide them with the necessary counselling or academic guidance.

The prototype shows that a simplified AI-based monitoring system, developed on existing institutional data, and lacking sophisticated predictive modeling, may prove useful to proactive academic intervention. The system helps decrease dropout rates and enhance student retention in higher education organizations because it allows tracking students and making informed decisions based on the data.

B. Future Work

Although the existing implementation meets its major goals, additional advances have the potential to increase the scalability, automation of the system, and impact in the institution with the specified technology stack.

The development directions of the future are:

- Implementation of the system in cloud-based systems like AWS, Azure or GCP to allow centralized accessibility among departments. E.g. Auto email or SMS notification plug-in where mentors are notified every time a student falls under High-risk levels.
- Features extension in dashboard in Streamlit to offer more comprehensive academic trend graphics and performance summaries.
- Role-based monitoring functionality improvement, allowing the organization of access to administrative information and data control.
- Optimization of rule-based thresholds with feedback on rules and real academic data in order to achieve better classification results with time.
- Periodic monitoring cycles to enable continuous risk assessment as the new student information is uploaded.

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REFERENCES

[1] A. L Bowers, R. Sprott and M. Taff, Do We Know Who Will Drop Out? A Review of the Predictors of Dropping Out of High School, Teachers College Record, vol. 115, no.10, pp. 1 40, 2013.

[2] M. Dekker, M. Pechenizkiy and J. Vleeshouwers, Predicting Students Drop Out: A Case Study, Proc. International Conference on Educational Data Mining (EDM), 2009, pp. 41 50.

[3] S. Kotsiantis, C. Pierrakeas and P. Pintelas, "Performance in Distance Learning: Machine learning to predict the performance of students with the use of machine learning techniques (18), no. 5, pp. 411-426, 2004.

[4] Romero and S. Ventura, E.D.M.A. IEEE Transactions on Systems, Man, Cybernetics, Part C (Applications and Reviews), vol. 40, no. 6, pp. 601618, Nov. 2010.

[5] R. S. J. d. Baker and K. Yacef, The State of Educational Data Mining in 2009: A Review and Future Visions, Journal of Educational Data Mining vol. 1, no. 1, pp. 317, 2009.

[6] H. Aldowah, H. Al-Samarraie, and W. M. Fauzy, Educational Data Mining and Learning Analytics in 21 st Century Higher Education: Review and Synthesis, Telematics and Informatics, vol. 37, pp. 13 49, 2019.

[7] E. Tinto,Dropout from Higher Education: ATheoreticalSynthesis ofRecentResearch, Review of Educational Research, vol. 45, no. 1, pp. 89125, 1975.

[8] S. Helal, J. Li, L. Liu, E. Ebrahimie, and S. Dawson, "Student Academic Performance by Subgroup Discovery: Identifying Key Factors of Academic Performance in Students," International Journal of Data Science and Analytics, vol. 7, no. 3, pp. 227245, 2019.

[9] J. L. Beck and J. Mostow, How Who Should Practice Using Learning Decomposition to estimate the effectiveness of various kinds of practice by various kinds of student, in Proc. International Conference on Intelligent Tutoring Systems, 2008, pp. 353362.

[10] A. Peña-Ayala, Educational Data Mining: A Survey and a Data Mining-Based Analysis of Recent Works, Expert Systems with Applications, vol. 41, no. 4, pp. 14321462, 2014.

[11] D. Kabkchieva, Predicting Student Performance using Data Mining method to classify data, Cybernetics and Information Technologies, vol. 13, no. 1, pp. 6172, 2013.

[12] S. B. Kotsiantis, Use of Machine Learning Techniques to Educational Proposes: A Decision

Support System to Forecast the grades of students, Artificial Intelligence Review, vol. 37, no. 4, pp. 331344, 2012.

[13] J. Xu and B. Jaggars, Performance Gaps Between Online and Face-to-Face Courses: Differences byTypes of Students and Academic Subject Areas, Journal of Higher Education,vol. 85, no. 5, pp. 633 659, 2014.

[14] UNESCO, Global Education Montoring Report 2023: Technlogy in Education A Taol on Whose Terms? 2023, United Nations Educational, Scientific and Cultural Organization, Paris, France.

[15] Machine Learning T. M. Mitchell. New York, NY, USA: McGraw-Hill, 1997.